

# Document Intelligence Platform

- AI Engineer Internship Case Study
- Dynamic extraction for Invoice, Balance Sheet, Profit & Loss, and Cash Flow Statement.
- OCR • 2-D layout • dynamic fields • grounded LLM • validation • SQLite • REST API • dashboard

## Problem and design goal

- Avoid vendor/template-specific parsers and growing regex/if-else rules.
- Handle different labels, layouts, table positions, headers, footers, and wrapped descriptions.
- Separate document processing status from financial validation status.

# Dynamic processing pipeline

- Upload → file validation → native text/OCR → 2-D layout extraction → dynamic semantic extraction → evidence verification → financial validation → persistence → dashboard/API
- OCR retains word coordinates so invoice columns are not flattened into a single text stream.

# Layout-aware invoice extraction

- Infer Description / Quantity / Unit Price / Amount columns from actual header positions.
- Numbers inside product descriptions such as 3M 570, 12oz, and 100/BX remain in the description column.
- Wrapped description rows can be buffered and merged without stealing values from adjacent numeric columns.

## Dynamic fields and LLM grounding

- Canonical fields support required financial validation concepts.
- Unknown meaningful fields are preserved in `discovered_fields`.
- LLM receives the full OCR document, including header, right-side content, footer, and all pages.
- LLM values are accepted only when grounded in supplied OCR evidence.

# Invoice validation

- Total formula:  $\text{subtotal} + \text{tax} + \text{shipping/handling/other charges} - \text{discount} \approx \text{total}$ .
- Line-item  $\text{quantity} \times \text{unit price} \approx \text{amount}$ .
- Line-item  $\text{sum} \approx \text{subtotal}$ .
- Validation discrepancies are isolated in `validation.overall_status`; a successfully parsed document remains `processing_status = PASS`.

# Statement validation

- Balance Sheet: total capital/liabilities  $\approx$  total assets; component reconciliations where sufficient data exists.
- P&L;: assignment formulas for bank-style and generic statements.
- Cash Flow: operating + investing + financing + FX  $\approx$  net change; opening + net change + adjustments  $\approx$  closing.

## **API, frontend and persistence**

- FastAPI REST endpoints for process, list, detail, and health.
- SQLite persistence for processed results.
- Dashboard shows extracted fields, evidence, discovered fields, line items, validation checks, and raw JSON.
- Missing values remain null rather than fabricated.

## Testing and limitation

- 31 tests passed, 1 skipped locally after the dynamic/layout/status changes.
- The supplied dataset contains real scanned Balance Sheets but not real source documents for Invoice/P&L;/Cash Flow.
- The batch2-0499 issue is covered by a 2-D column extraction regression test.

# Deliverables

- Dynamic source code and layout-aware OCR
- Architecture diagram and technical audit
- Solution presentation (PPTX and PDF)
- Automated tests
- README with setup, API, validation, and deployment guidance